



Clustering Documents along Multiple Dimensions

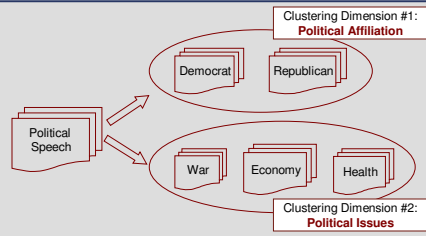
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Motivation

Many documents can be clustered along multiple **dimensions**



But traditional text clustering algorithms are designed to produce a single clustering, typically a topic-based clustering

Goal

Design a **multi-clustering** algorithm that can

- discover the important dimensions of a set of text documents
- produce a clustering along each such dimension

• Important from the viewpoint of exploratory data analysis

- helps a user discover how a data sample can be clustered

Problem Formulation

Input: data sample $X_n \equiv \{x_1, x_2, \dots, x_n\}$
 z : number of clusterings to be produced
 k_n : number of clusters within each clustering $K^{(m)}$

Output: a **multi-clustering** solution, i.e., a set $K \equiv \{K^{(1)}, \dots, K^{(z)}\}$ of z clusterings, where each clustering $\{C_1^{(m)}, \dots, C_{k_n}^{(m)}\}$ consists of a finite set of k_n clusters

How to determine whether solution K_1 is better than solution K_2 ?

- employs **three constraints** for a pairwise preference relation
- Quality:** K_1 is better than K_2 if K_1 has a higher quality than K_2 according to a clustering quality preference function
- Distinctivity:** K_1 is better than K_2 if the clusterings in K_1 are more distinctively different than those in K_2 according to a clustering dissimilarity preference function
- Parsimony:** K_1 is better than K_2 if K_1 is more resistant to overfitting than K_2 according to parsimony constraints

Approach

Proposed the Probabilistic Multi-Clustering (PMC) model

- a **generative, knowledge-lean** approach to multi-clustering that
 - supports Bayesian decision rules (e.g., minimum probability of error rules) for the assignment of data points to clusters
 - assumes a **single** feature space composed of unigrams only, thus obviating the need to manually identify the relevant features for each dimension
- extends the **Diagonal Gaussian Mixture Model (DGMM)** to produce multiple clusterings

Diagonal Gaussian Mixture Model

• can be used to generate a **single** clustering of a data sample

Definition: A Gaussian Mixture Model (GMM) is a weighted sum of k component Gaussian densities:

$$p(x|\Theta) = \sum_{j=1}^k p(C_j|\gamma) p(x|C_j, \mu_j, \sigma_j)$$

Labels: d-dimensional feature vector, number of clusters, k-dimensional vector denoting cluster relevance, mean of jth Gaussian, Std. dev. of jth Gaussian, set of model parameters, mixture weights, component Gaussian densities

Definition: A Diagonal Gaussian Mixture Model (DGMM) is a GMM where the covariance matrix of each component density is diagonal.

Generative story (2 steps):

- A cluster C_j is generated with probability $p(C_j|\gamma^*)$ for some γ^*
- Given C_j , a point x is generated with probability $p(x|C_j, \mu_j, \sigma_j)$

Parameter estimation

- Θ is composed of $3k$ parameters, with 3 parameters per component
 - component j has parameters μ_j, σ_j and γ_j
- Θ can be computed by maximizing the log likelihood using EM, Generalized GM, or gradient descent

Clustering

- the learned Θ can be used to induce a soft clustering of documents
- a hard clustering can be derived from the soft clustering by assigning each point x to the most likely cluster C_j given x

Multi-Clustering DGMM (MDGMM)

- extends the DGMM to produce multiple clusterings based on three multi-clustering criteria: quality, distinctivity, and parsimony

A naïve MDGMM for generating z clusterings:

- make z copies of the given data sample
- employ z DGMMs to independently induce z clusterings
 - each DGMM generates 1 copy of the data & induces 1 clustering
- estimate the $3kz$ parameters by maximizing the log likelihood of jointly generating the z copies of the data sample

Weakness: Maximizing data likelihood ensures quality, but not distinctivity and parsimony

Solution: Define a prior for the naïve MDGMM that incorporates

- distinctivity** constraints to favor solutions with dissimilar clusterings
- parsimony** constraints to regularize the parameter space

Implementing distinctivity constraints

- Define soft-clustering cluster-location similarity preference function

$$\phi_{\mu_j}^{(m)} = \left(\frac{1}{m-1} \sum_{u=1}^{m-1} \frac{1}{k_u} \sum_{q=1}^{k_u} (\mu_j^m \cdot \mu_q^u)^2 \right)^2$$

for $m = 2, 3, \dots, z$ and $\phi_{\mu_j}^{(1)} = 1$

- computes the average similarity between the location of cluster j in clustering m and the location of all the clusterings induced thus far
- smaller $\phi_{\mu_j}^{(m)}$ corresponds to more dissimilar cluster locations

Parsimony constraints as L2-regularization functions

Since the m th DGMM in the MDGMM has three sets of parameters $(\mu_j^{(m)}, \sigma_j^{(m)}, \gamma_j^{(m)})$, we define a regularizer for each set of parameters

$$\eta_{\mu,j}^{(m)} = \|\mu_j^{(m)}\|^2, \eta_{\sigma,j}^{(m)} = \|\sigma_j^{(m)}\|^2, \eta_{\gamma}^{(m)} = \|\gamma^{(m)}\|^2$$

Combining the two types of constraints as a prior

- Define clustering solution preference function $V(\Theta)$ such that:

$$V(\Theta) \equiv \sum_{m=1}^z \sum_{j=1}^{k_m} V_j^{(m)} \quad \text{where} \quad V_j^{(m)} = \lambda_{\mu,j}^{\mu} \phi_{\mu,j}^{(m)} + \lambda_{\sigma,j}^{\sigma} \eta_{\sigma,j}^{(m)} + \lambda_{\gamma}^{\gamma} \eta_{\gamma}^{(m)}$$

and $\lambda_{\mu,j}^{\mu}, \lambda_{\sigma,j}^{\sigma}, \lambda_{\gamma}^{\gamma}$ are non-negative real numbers determined based on user's prior knowledge of the importance of each constraint

- Define Bayesian parameter prior p_{Θ} as a function of $V(\Theta)$ such that:

$$p_{\Theta}(\Theta) = \frac{1}{Z} \exp \left(\frac{-V(\Theta)}{2\sigma_p^2} \right) \quad \text{where } \sigma_p \text{ controls the intensity of the prior}$$

- Lowering σ_p increases the effect of distinctivity and parsimony

Modifying the objective function

- Incorporate prior p_{Θ} into the objective used by the naïve MDGMM
- Seek a MAP estimate of the resulting objective function

Parameter estimation

- employs BFGS within a 5-step multi-stage estimation algorithm
- Idea:** generate a series of progressively more complicated estimation problems using the results of the previously solved problem as an initial guess for the next more complicated problem

Evaluation

Goal: evaluate the effectiveness of our PMC model in producing multiple clusterings along important dimensions of a text dataset

Five text datasets: each has multiple 2-way clustering dimensions

Dataset	Description	Dimensions (and Cluster Labels)
BOOK	2000 book reviews	Sentiment (Positive vs. Negative) Subjectivity (Subjective vs. Objective)
DVD	2000 DVD reviews	Sentiment (Positive vs. Negative) Subjectivity (Subjective vs. Objective)
MIX	4000 reviews from BOOK and DVD	Sentiment (Positive vs. Negative) Subjectivity (Subjective vs. Objective) Topic (Book vs. DVD)
MAN	1000 blog posts	Gender (Male vs. Female) Age (Old vs. Young)
POL	2000 political articles	Pol. Affiliation (Dem. vs. Republican) Policy (Foreign vs. Domestic)

Evaluation metric: accuracy

Three Baseline Multi-Clustering Algorithms

- Meta clustering (Caruana et al., 2006): unsupervised
 - run k -means multiple times, each time with a random selection of seeds and a random weighting of features
 - report best accuracy achieved over 100 runs for each dimension
- GMM-EM: unsupervised
 - run GMM z times, each time using EM for parameter estimation and selecting seeds randomly, to produce z clusterings, with $z=5$
 - find the 1-to-1 mapping between the z proposal clusterings and the gold standard clusterings that yields the highest accuracy
 - report average accuracy of a proposal clustering against the mapped gold clustering over 3 runs
- Davidson and Qi's (2007) algorithm: semi-supervised
 - given a dataset with two clusterings, it assumes one of them is supplied (by a human) as input and aims to produce the other
 - for each dataset with dimensions i and j , we feed it with the gold clustering for dimension i and measure the quality of the clustering it proposes against the gold clustering for dimension j
 - repeat the experiment by reversing the roles of i and j

Results

	BOOK		DVD	
System	Sent.	Subj.	Sent.	Subj.
Meta clustering	50.8	51.2	53.9	71.0
GMM-EM	50.3	67.4	53.3	72.2
Davidson & Qi	50.9	55.7	51.2	59.6
PMC	59.1	61.5	57.1	61.7

	MAN		POL	
System	Gender	Age	Affil.	Policy
Meta clustering	51.2	53.6	59.4	58.8
GMM-EM	51.8	52.8	55.1	57.1
Davidson & Qi	50.5	50.9	50.5	51.2
PMC	63.0	56.5	64.6	62.2

	MIX		
System	Topic	Sent.	Subj.
Meta clustering	50.2	50.2	58.6
GMM-EM	50.7	52.1	69.5
Davidson & Qi	50.5	51.2	57.8
PMC	64.8	60.5	63.5

Discussion

- Meta clustering
 - performs poorly though best results are reported
- GMM-EM
 - performs well only along Subjectivity
 - poor performance can be attributed to its producing similar clusterings for each dataset
- Davidson and Qi's algorithm
 - does not perform better than other baselines despite its being semi-supervised
- Our PMC model
 - achieves best result for 8 of the 11 dimensions
 - achieves stable performance across dimensions

Output produced by PMC for the MIX dataset

- PMC discovers 4 clusterings: $K^{(1)}, K^{(2)}, K^{(3)}, K^{(4)}$
- each clustering consists of two clusters: C_1, C_2
- top words are selected on the basis of their relative frequency of occurrence in each cluster

MIX			
$K^{(1)}$	$K^{(2)}$	$K^{(3)}$	$K^{(4)}$
C_1	C_1	C_1	C_1
text	watched	mind	relationship
knowledge	music	novel	lives
death	bought	readers	human
national	episode	users	wonderful
information	version	reader	women
using	wanted	nature	age
C_2	C_2	C_2	C_2
films	readers	young	disappointed
scene	national	features	caught
reason	ask	military	action
ending	facts	disc	novel
script	reader	video	mystery
absolutely	destruction	screen	isn
Topic	Subjectivity	Topic	Sentiment